

AI Banking Assistant - AWS Verification Checklist

Recommended Region: us-east-1

1. DynamoDB - Contracts Table (us-east-1)

- ☐ **Table exists** Table named "Contracts" is created in DynamoDB console
- ☐ **Partition key** contract_id (String type)
- ☐ **Test item inserted** Insert sample record with all required fields
- ☐ **Billing mode** On-demand or provisioned capacity configured

Sample test item:

```
{
  "contract_id": "C123",
  "amount": 500000,
  "interest_rate": 0.05,
  "duration": "5 years"
}
```

2. Amazon Bedrock - Model Access (us-east-1)

- ☐ **Model access enabled** anthropic.claude-3-haiku-20240307-v1:0 enabled in Bedrock console
- ☐ **Access requested** If not enabled: Bedrock > Model access > Manage model access
- ☐ **Invoke permissions** IAM role/user has bedrock:InvokeModel permission

Supported regions for Claude 3 Haiku: us-east-1, us-west-2, eu-west-1, ap-northeast-1

3. IAM Permissions (Global)

- ☐ **DynamoDB access** dynamodb:GetItem on
arn:aws:dynamodb:us-east-1:{account}:table/Contracts
- ☐ **Bedrock access** bedrock:InvokeModel on the Claude 3 Haiku foundation model ARN
- ☐ **Credentials configured** aws configure or environment variables set with valid credentials

Minimal IAM Policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "dynamodb:GetItem",
      "Resource": "arn:aws:dynamodb:us-east-1:*:table/Contracts"
    },
    {
      "Effect": "Allow",
      "Action": "bedrock:InvokeModel",
      "Resource": "arn:aws:bedrock:us-east-1::foundation-model/anthropic.claude-3-haiku-20240307-v1:0"
    }
  ]
}
```

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4. Local Environment Setup

- | | |
|--|--|
| ☐ AWS credentials valid | <code>aws sts get-caller-identity</code> returns valid identity |
| ☐ Region configured | <code>AWS_DEFAULT_REGION=us-east-1</code> or set in <code>~/.aws/config</code> |
| ☐ Dependencies installed | <code>pip install -r requirements.txt</code> completes successfully |
| ☐ App starts | <code>uvicorn main:app --reload</code> starts without errors on port 8000 |

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5. End-to-End Verification Tests

- ☐ **No contract_id** Returns prompt message, status AUTO

Command:

```
curl -X POST http://localhost:8000/chat \  
  -H "Content-Type: application/json" \  
  -d '{"message": "Hello"}'
```

Expected: message contains "provide a contract_id", status = "AUTO"

- ☐ **Valid contract_id** Returns summary, status AUTO

Command:

```
curl -X POST http://localhost:8000/chat \  
  -H "Content-Type: application/json" \  
  -d '{"message": "Show my contract", "contract_id": "C123"}'
```

Expected: contract_summary is populated, status = "AUTO"

- ☐ **Non-existent contract** Returns escalation message

Command:

```
curl -X POST http://localhost:8000/chat \  
  -H "Content-Type: application/json" \  
  -d '{"message": "Show contract", "contract_id": "DOESNOTEXIST"}'
```

Expected: message = "Contract not found, escalating to support", status = "ESCALATE"

- ☐ **Invalid request** Returns 422 validation error

Command:

```
curl -X POST http://localhost:8000/chat \  
  -H "Content-Type: application/json" \  
  -d '{"contract_id": "C123"}'
```

Expected: HTTP 422 with validation error (missing message field)

6. Expected HTTP Response Codes Summary

Scenario	HTTP Code	Status Field
No contract_id provided	200	AUTO
Contract found + summary OK	200	AUTO
Contract not found	200	ESCALATE
DynamoDB connection error	502	ESCALATE
Bedrock error/timeout	200	ESCALATE
Invalid request body	422	N/A
Unexpected server error	500	ESCALATE

Tip: Use us-east-1 for all services to keep DynamoDB and Bedrock co-located for lowest latency.